# Reliable Annotation and Evaluation of Malaysian Malay-English Code-Mixed Sentiment Data: A Critical Review of an Unresolved Problem

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***Abstract***

*Malaysian social media users mix English and Malay inside single messages, producing the contact varieties known as Bahasa Rojak and Manglish. Sentiment analysis of this text is usually treated as a modelling problem. This review indicates that the more limiting problem lies earlier, in the sentiment labels themselves and in the evidence offered that those labels are sound. The review covers annotation practice and reliability evaluation for Malaysian Malay-English code-mixed sentiment data. Comparative benchmarking of sentiment classifiers is outside its scope. Three contributions follow. First, the annotation strategies used for low-resource code-mixed text are organised into three tiers, and each tier is shown to improve either label fidelity or scalability by giving up the other. Second, the two causes of neutral-label skew are separated, and the four regional corpora are audited against what a reliability claim requires. No corpus reports the full set of elements, the highest reported coefficient measures a narrower task than the resource is used for, and one manually annotated corpus reports no agreement statistic at all. Third, the class imbalance that affects the labels is shown to be the same condition under which the standard agreement statistic becomes difficult to interpret. The review closes with a research agenda covering minimum reporting standards, agreement measurement under class imbalance, and the untested assumption that annotators of different types make different errors.*

***Keywords:*** *sentiment analysis, code-mixed text, Bahasa Rojak, annotation reliability, inter-annotator agreement*

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## 1. Introduction

Sentiment analysis, also known as opinion mining, is the task of automatically deciding whether a piece of text expresses a positive, negative or neutral opinion [Poria et al. (2023)]. Organisations use it to measure public opinion at scale, in applications that range from brand monitoring to political and policy analysis [Poria et al. (2023)], [Nazir et al. (2022)]. Most of these applications were built and tested on text written in one standard language. Malaysian social media text does not meet that condition.

Malaysia is a multilingual and multi-ethnic country in which around 140 languages are spoken, and code-mixing between a speaker's first language, English and Malay is already widespread in everyday use [Coluzzi (2017)]. This carries over to social media, where users often switch between languages inside a single message instead of writing in one language at a time [Treffers-Daller et al. (2022)], [AlAfnan et al. (2025)]. English and Malay are the pair that meets most often in written form, and the mixed varieties that result are known locally as *Bahasa Rojak* and *Manglish* [AlAfnan et al. (2025)]. This kind of writing is common enough on X that both of the Malaysian sentiment corpora examined in this review were collected from that platform [Maskat et al. (2024)], [Mohd Shamsuddin et al. (2024)].

Manglish is sometimes described as a variety that weakens standard English and Malay rather than as a language form in its own right [AlAfnan et al. (2025)]. Its structure, however, is not random. Treffers-Daller et al. [Treffers-Daller et al. (2022)] and AlAfnan et al. [AlAfnan et al. (2025)] show that Malay-English switching follows describable syntactic and pragmatic patterns, and that speakers use it deliberately to communicate efficiently, signal identity and build rapport. The difficulty this text creates for natural language processing (NLP) is therefore not that it is written carelessly. It is that the rules Malaysian writers follow are not the rules that standard NLP tools were built to handle, so writing that is systematic is processed as noise [Winata et al. (2023)], [Abu Bakar et al. (2020)].

When a sentiment model performs poorly on this material, the explanation usually given is architectural. The argument is that the model cannot handle code-switched input, and that a larger or more regionally adapted model would perform better on such data. This argument is reasonable, and part of the literature supports it. Khanuja et al. [Khanuja et al. (2020)] report that multilingual models fine-tuned on code-switched data give the best results on their benchmark, and Adilazuarda et al. [Adilazuarda et al. (2022)] show that a model's robustness to code-mixing depends on how well each language is represented during pre-training. The same literature, however, also reports the limits of this route. Winata et al. [Winata et al. (2021)] find that multilingual pre-training on its own does not guarantee good representations of code-switched text, and Conneau et al. [Conneau et al. (2020)] describe a "curse of multilinguality" in which finite model capacity, once spread across many languages, dilutes per-language quality. Model choice is therefore part of the problem, but it is not all of it.

A second part of the problem sits earlier in the process. Before a model can be trained or evaluated, every message must be read and given a sentiment label, either

by a person or by a program. This step is called annotation, and the labels it produces form the supervision signal that a model learns from and is measured against [Artstein & Poesio (2008)]. Annotation quality matters because label errors do not stay inside the dataset. They are learned by the model during training [Frénay & Verleysen (2014)], and they also distort the test set that is used to judge how good the model is [Northcutt et al. (2021)], [Bowman & Dahl (2021)]. This review indicates that for Malaysian Malay-English code-mixed sentiment data, the more limiting problem at present is the annotation label itself, together with the evidence offered that the label is sound.

Two observations support this reading, and both recur in the sections that follow. The first concerns the corpora themselves. Four Malay-English sentiment resources are currently available, and none of them is at the same time large, balanced across sentiment classes, drawn from casual social media, and validated by human annotators [Mohd Shamsuddin et al. (2024)], [Kasmuri & Basiron (2019)], [Kong et al. (2023)], [Romadhona et al. (2022)]. Each resource obtains some of these properties by giving up others. The largest general-domain resource, MESocSentiment, shows the cost most clearly: its 19,714 posts are labelled 78.23% neutral, 16.33% positive and 5.44% negative [Mohd Shamsuddin et al. (2024)]. Two pieces of evidence indicate that this distribution comes from the labelling method rather than from Malaysian discourse. First, the corpus was collected under the "#Malaysia" keyword across sports, politics and the arts, which are not topics on which social media users are usually neutral [Mohd Shamsuddin et al. (2024)]. Second, a comparable corpus of Malaysian tweets labelled by human annotators is only 23.6% neutral [Kong et al. (2023)]. The neutral share differs by a factor of more than three between two corpora of the same kind of text.

The second observation concerns gaps in the reporting of annotation reliability. Reliability is normally demonstrated by having independent annotators label the same text and measuring how often they agree, after subtracting the agreement that chance alone would produce [Artstein & Poesio (2008)]. Among the four regional resources, this evidence is thin. MESocSentiment reports no agreement statistic, which is unsurprising because it has no human annotation stage [Mohd Shamsuddin et al. (2024)]. MyCovid-Senti also reports none, although it was annotated by three people working from a written guideline [Kong et al. (2023)]. The two resources that do report a coefficient measured a binary judgement rather than the three-class polarity task for which the resources are actually used. Kasmuri and Basiron [Kasmuri & Basiron (2019)] report $\kappa = 0.83$ for an opinionated-versus-factual decision. Romadhona et al. [Romadhona et al. (2022)] report $\kappa = 0.77$ for a two-way positive-or-negative judgement on stock-market posts, covering one subset of their corpus only. The field therefore lacks not only reliable annotations but also established practices for validating their reliability.

These two weaknesses are usually treated as separate problems, but they are connected. The class imbalance that affects the labels is also the condition under which the standard agreement statistic becomes difficult to interpret. It is this combined weakness of annotation and evaluation, rather than either weakness on its own, that helps explain why the problem has proved persistent.

This review examines one question: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data remain unresolved. Linguistic description, model architecture and application domain are discussed only where they bear on that question. Comparative benchmarking of sentiment classifiers is outside the scope, as is sentiment analysis as a general field, because both have already been surveyed in detail by Poria et al. [Poria et al. (2023)], Nazir et al. [Nazir et al. (2022)] and Selvaraju et al. [Selvaraju et al. (2024)].

It is also useful to state what this review does not do. It does not propose, implement or evaluate a system, and it does not report the author's own experimental work. Its contribution is a diagnosis of why the problem has resisted solution, together with a set of directions for future research. Where the analysis finds that a question has not yet been addressed in the literature, it identifies the question and leaves it open for future work.

This review makes three contributions:

1. It organises the annotation strategies used for low-resource code-mixed text into three tiers, and shows that each tier improves either label fidelity or scalability by giving up the other.
2. It separates the two different causes of neutral-label skew, and audits what the four regional corpora report about their own reliability.
3. It shows how the linguistic, annotation and measurement challenges interact rather than simply occurring together, and sets out a research agenda that follows from that interaction.

## 2. Review Methodology

The literature reviewed here was identified through structured searches of four databases: the ACL Anthology, IEEE Xplore, Scopus and arXiv. Three groups of search terms were combined. The first group covered the language variety and included the terms code-mixed, code-switching, Manglish, Bahasa Rojak and Malay-English. The second group covered the task and included sentiment analysis, polarity and opinion mining. The third group covered the object of interest and included corpus, annotation, inter-annotator agreement and label quality.

The search window was 2019 to 2026. It was extended backwards by following citations from the corpus papers retrieved in the initial search. The date filter was not applied to work on inter-rater reliability.

Preprints were treated with caution. A paper from arXiv was included only when no peer-reviewed version could be found and the preprint had a traceable submission record. Where a peer-reviewed version of a preprint exists, that version is the one cited.

Four criteria were used to determine inclusion. A study was included only if it met all four:

1. It addressed Malaysian or closely comparable Malay-English code-mixed text, or supplied annotation or reliability methodology that can be applied to such text;

2. It reported enough procedural detail for its annotation or evaluation practice to be assessed;
3. It was peer-reviewed, or was a preprint with a traceable venue; and
4. It contained an annotation or reliability component.

The fourth criterion is the one that keeps the scope narrow. Studies that only benchmark classifiers against an existing corpus were excluded, however relevant their results may be to Malaysian sentiment analysis in general. Such studies treat the quality of the existing labels as given, whereas this review asks whether that quality can be established in the first place.

## 3. Why Malaysian Code-Mixed Text Is Difficult to Annotate

Three properties of Malaysian code-mixed writing affect the annotation task directly. All three have already been described in the sociolinguistic literature on Malay-English contact [Treffers-Daller et al. (2022)], [AlAfnan et al. (2025)], [Muysken (2013)]. What this section adds is not the description but the consequence that each property carries for a person or a program that has been asked to assign a sentiment label. Each property is therefore followed by a short statement of that consequence, and is not developed further.

### 3.1 Switching happens inside the sentence, and the switch point gives no structural cue

Malay-English switching happens mostly inside the clause rather than between clauses. Treffers-Daller et al. [Treffers-Daller et al. (2022)] analysed 1,044 switched utterances and found that insertion, in which a content word or phrase from one language is embedded in a stretch of the other, accounted for 33.0% of switches. Congruent lexicalisation, in which the grammar and vocabulary of both languages operate within a shared word order, accounted for a further 30.7%. Clause-level alternation accounted for 24.8%. The authors explain this pattern typologically rather than stylistically. English and Malay share Subject-Verb-Object (SVO) word order, and Malay carries very little inflectional morphology, so a speaker can move between the two languages in mid-clause with almost no grammatical friction [Treffers-Daller et al. (2022)], [Muysken (2013)].

Two limits on this evidence should be noted. The corpus consists of spoken data from two teachers, so the exact proportions should not be transferred to social media text. However, the structural explanation is a property of the language pair rather than of the medium, so it applies wherever the two languages meet. A written example is the tweet "Does institut kanser offer free medication cheap chemotherapy", in which a Malay noun phrase, in Malay head-initial word order, sits inside an English question [Mohd Shamsuddin et al. (2024)].

The consequence for annotation is that the annotator cannot use a language boundary to locate the part of the message that carries the sentiment. In monolingual annotation, the phrase carrying the opinion can usually be identified before it is judged. Here, identifying it is already an act of interpretation. Two annotators may therefore assign different labels not because they read the sentiment differently, but

because they are reading different parts of the message. Disagreement about which span to judge comes before, and is separate from, disagreement about polarity.

## References

Abu Bakar, M. F. R., Idris, N., Shuib, L., & Khamis, N. (2020). Sentiment analysis of noisy Malay text: State of art, challenges and future work. *IEEE Access, 8*, 24687–24696. https://doi.org/10.1109/ACCESS.2020.2968955

Adilazuarda, M. F., Cahyawijaya, S., Winata, G. I., Fung, P., & Purwarianti, A. (2022). IndoRobusta: Towards robustness against diverse code-mixed Indonesian local languages. *Proceedings of the First Workshop on Scaling Up Multilingual Evaluation (SUMEval)*, 25–34. https://aclanthology.org/2022.sumeval-1.5/

AlAfnan, M. A., MohdZuki, S. F., & AlAfnan, S. M. (2025). Blending identities: Exploring the linguistic, cultural, and pragmatic dimensions of Manglish in the Malaysian society. *Studies in Media and Communication, 13*(3). https://doi.org/10.11114/smc.v13i3.7654

Artstein, R., & Poesio, M. (2008). Survey article: Inter-coder agreement for computational linguistics. *Computational Linguistics, 34*(4), 555–596. https://doi.org/10.1162/coli.07-034-R2

Bowman, S. R., & Dahl, G. (2021). What will it take to fix benchmarking in natural language understanding? *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 4843–4855. https://doi.org/10.18653/v1/2021.naacl-main.385

Coluzzi, P. (2017). Language planning for Malay in Malaysia: A case of failure or success? *International Journal of the Sociology of Language, 2017*(244), 17–38. https://doi.org/10.1515/ijsl-2016-0055

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 8440–8451. https://doi.org/10.18653/v1/2020.acl-main.747

Frénay, B., & Verleysen, M. (2014). Classification in the presence of label noise: A survey. *IEEE Transactions on Neural Networks and Learning Systems, 25*(5), 845–869. https://doi.org/10.1109/TNNLS.2013.2292894

Kasmuri, E., & Basiron, H. (2019). Building a Malay-English code-switching subjectivity corpus for sentiment analysis. *International Journal of Advanced Soft Computing and Its Applications, 11*(1), 112–130.

Khanuja, S., Dandapat, S., Srinivasan, A., Sitaram, S., & Choudhury, M. (2020). GLUECoS: An evaluation benchmark for code-switched NLP. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 3575–3585. https://doi.org/10.18653/v1/2020.acl-main.329

Kong, J. T. H., Juwono, F. H., Ngu, I. Y., Nugraha, I. G. D., Maraden, Y., & Wong, W. K. (2023). A mixed Malay-English language COVID-19 Twitter dataset: A sentiment analysis. *Big Data and Cognitive Computing, 7*(2), 61. https://doi.org/10.3390/bdcc7020061

Maskat, R., Azman, N. A., Nulizairos, N. S. S., Zahidin, N. A., Mahadi, A. H., Norshamsul, S. R., Mohd Sharif, M. M., & Mahdin, H. (2024). A bi-annotated Malay-English code-switching (Manglish) dataset of X posts for biological gender identification and authorship attribution. *Data in Brief, 52*, 110034. https://doi.org/10.1016/j.dib.2024.110034

Mohd Shamsuddin, A. M., Samson Juan, S., Chua, S., & Bramantoro, A. (2024). Semi-automatic sentiment identification for Malay-English code-switched data. *Journal of Advanced Research Design, 123*(1), 198–212. https://doi.org/10.37934/ard.123.1.198212

Muysken, P. (2013). Language contact outcomes as the result of bilingual optimization strategies. *Bilingualism: Language and Cognition, 16*(4), 709–730. https://doi.org/10.1017/S1366728912000727

Nazir, A., Rao, Y., Wu, L., & Sun, L. (2022). Issues and challenges of aspect-based sentiment analysis: A comprehensive survey. *IEEE Transactions on Affective Computing, 13*(2), 845–863. https://doi.org/10.1109/TAFFC.2020.2970399

Northcutt, C. G., Athalye, A., & Mueller, J. (2021). Pervasive label errors in test sets destabilize machine learning benchmarks. *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS): Datasets and Benchmarks Track*.

Poria, S., Hazarika, D., Majumder, N., & Mihalcea, R. (2023). Beneath the tip of the iceberg: Current challenges and new directions in sentiment analysis research. *IEEE Transactions on Affective Computing, 14*(1), 108–132. https://doi.org/10.1109/TAFFC.2020.3038167

Romadhona, N. P., Lu, S.-E., Lu, B.-H., & Tsai, R. T.-H. (2022). BRCC and SentiBahasaRojak: The first Bahasa Rojak corpus for pretraining and sentiment analysis dataset. *Proceedings of the 29th International Conference on Computational Linguistics (COLING)*, 4418–4428. https://aclanthology.org/2022.coling-1.389/

Selvaraju, S., Sjarif, N. N. A., & Anuar, M. S. M. (2024). Sentiment analysis in code mixing Malay text: A review. *2024 5th International Conference on Artificial Intelligence and Data Sciences (AiDAS)*, 280–285. https://doi.org/10.1109/AiDAS63860.2024.10730298

Treffers-Daller, J., Majid, S., Thai, Y. N., & Flynn, N. (2022). Explaining the diversity in Malay-English code-switching patterns: The contribution of typological similarity and bilingual optimization strategies. *Languages, 7*(4), 299. https://doi.org/10.3390/languages7040299

Winata, G. I., Aji, A. F., Yong, Z. X., & Solorio, T. (2023). The decades progress on code-switching research in NLP: A systematic survey on trends and challenges. *Findings of the Association for Computational Linguistics: ACL 2023*, 2936–2978. https://doi.org/10.18653/v1/2023.findings-acl.185

Winata, G. I., Cahyawijaya, S., Liu, Z., Lin, Z., Madotto, A., & Fung, P. (2021). Are multilingual models effective in code-switching? *Proceedings of the Fifth Workshop on Computational Approaches to Linguistic Code-Switching*, 142–153. https://doi.org/10.18653/v1/2021.calcs-1.20